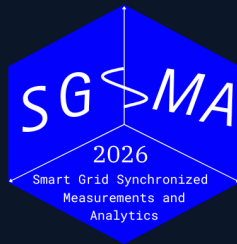


5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Date

June 1-4, 2026

Venue

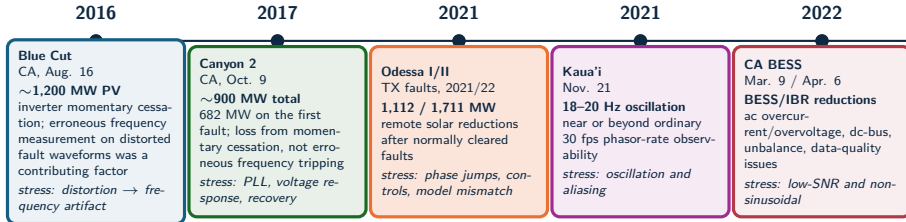
Santiago, Chile



Motivation

Why frequency estimation in low-inertia IBR grids is hard

Field events define the benchmark stress model



Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC/WECC Blue Cut (2017); 900 MW Canyon 2 (2018); NERC Odessa I/II (2021, 2022); NERC/WECC CA BESS (2023); Dong et al., Kaua'i 18–20 Hz oscillations (arXiv:2301.05781)

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Technical contributions

OUT OF SCOPE

- not PMU certification
- not a universal leaderboard
- not hardware-latency testing

1. OpenFreqBench

reproducible stress-test platform

2. Metric-complete protocol

RMSE, peak/RoCoF error, T_{trip} , settling, CPU

3. IBR failure evidence

standard cases miss composite IBR failures

4. Extensible estimator contract

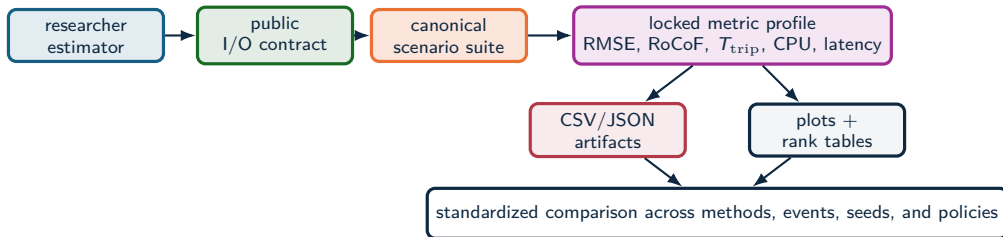
new methods enter the same scoring workflow

Contribution logic: explain estimator ranking shifts by objective, disturbance regime, and compute budget.

Benchmark & Platform

An open, reproducible estimator benchmark

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Principle: a submitted estimator becomes a repeatable benchmark entry under the same scenarios, seeds, metrics, and artifacts.

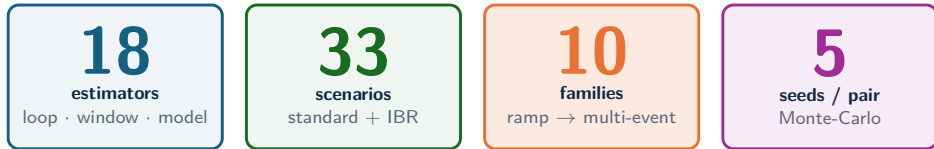
Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

Experiment scale



$18 \times 33 = 594$ estimator–scenario cells ■ $594 \times 5 = 2,970$
 Monte-Carlo records ■ 1 MHz truth → 10 kHz output

Central finding: no estimator dominates — winners change with event family, metric target, and compute budget.

Source: OpenFreqBench expanded scenario matrix

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL,
Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF,
LKF2.

Legacy baselines

TKEO and ZCD are retained as
comparison references outside the
recommended deployment set.

Window / spectral

IpDFT, TFT, ESPRIT, Prony,
MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different
latency and stress-sensitivity
positions. The comparison tracks
where each family wins, degrades,
and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

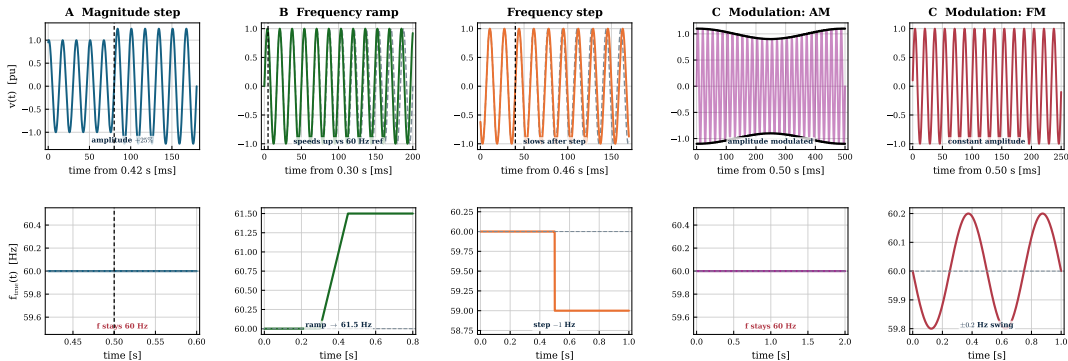
Scenarios

Standard and composite IBR stress cases

Standard stress scenarios

CORE SCENARIO SUITE

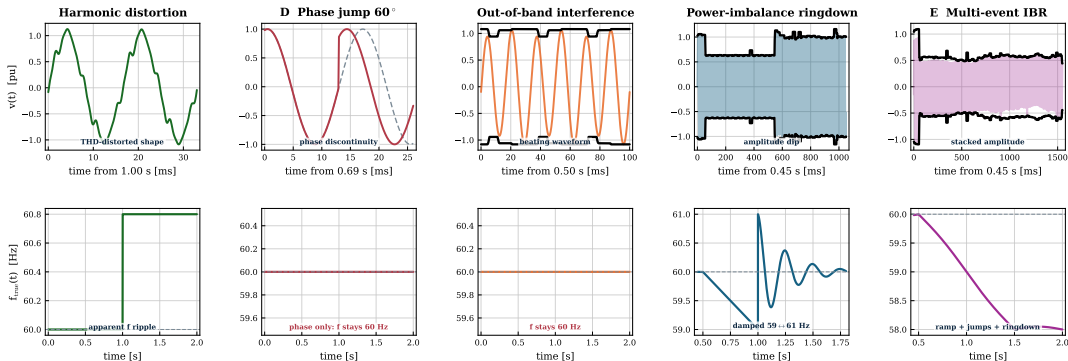
Scenario suite I — standard stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Composite IBR stress scenarios

IBR STRESS SUITE

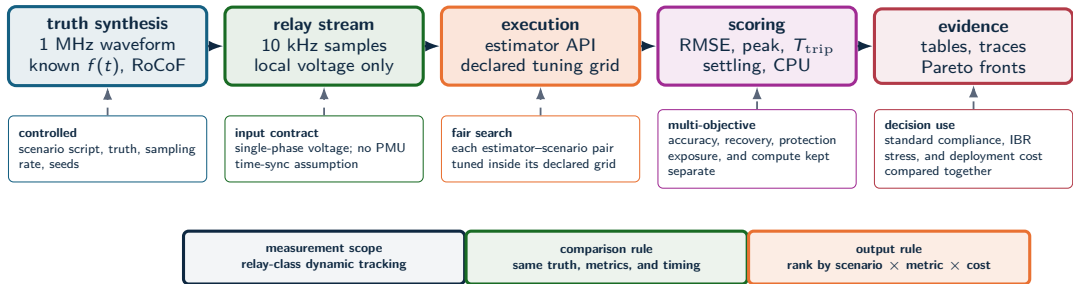
Scenario suite II — composite IBR stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



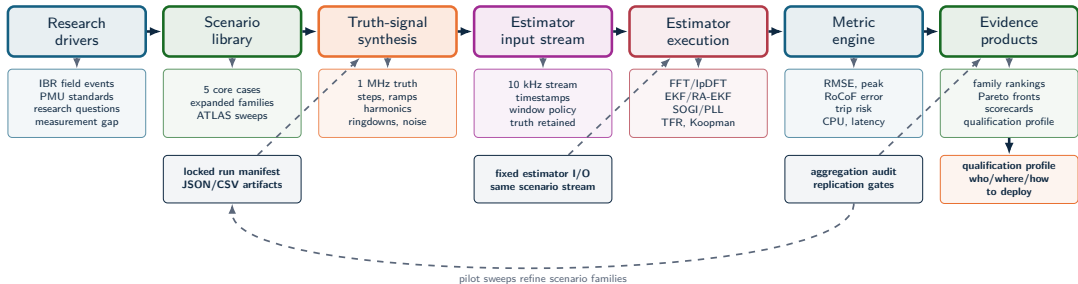
Metrics & Methodology

How every estimator is measured

Experimental method



Full benchmark workflow



The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Results

Winners depend on the question

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	0.006	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

LKF2 implementation after H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

What the heatmap shows

The magnitude step is easy for most methods. The ramp is the separator: windowed (IpDFT) and some filters can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

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UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028
PLL	0.033	1.47	0.227
IpDFT	0.000	14.4	0.114
EKF	0.000	37.1	0.064

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

LKF2 implementation after H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

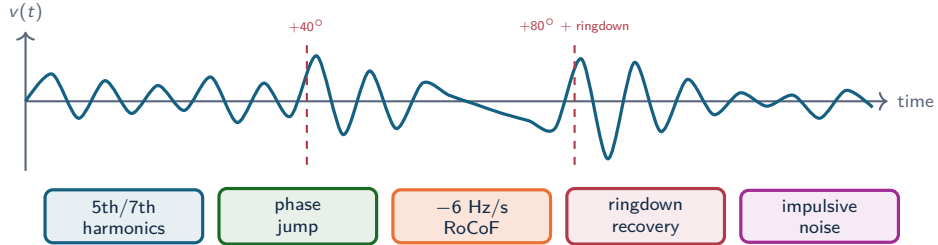
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Composite IBR sequence: why it is hardest



Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μ s]
UKF	1.09	0.78	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	0.78	11
SOGI-FLL	1.14	0.84	9
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	15 241
ZCD	> 400	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to ≈ 1.1 Hz RMSE; ZCD diverges catastrophically (hundreds of Hz, strongly seed-dependent). High accuracy is simply not available here.

Source: Monte-Carlo benchmark, $n = 5$ seeds -- composite multi-event scenario

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is $1000\times$ costlier; RA-EKF sits mid-pack but never diverges.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0 ; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz; windowed IpDFT 14 Hz diverges	Some filters show intermittent seed-dependent blow-ups
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000 $\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 $\mu\text{s}/\text{sample}$; Koopman $\sim 15,000$	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample [†]	Cost tier
SOGI-FLL	10 μ s	low
SOGI-PLL	11 μ s	low
EKF	12 μ s	low
RA-EKF	15 μs	low
TFT / IpDFT	12 μ s	low
ESPRIT	95 μ s	medium
Koopman	0.1–15 ms	high

[†] mean per-sample wall-clock in the Python harness (software proxy), $n = 5$;
Koopman/ESPRIT scale with event length.

Deployment interpretation

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

Key comparison

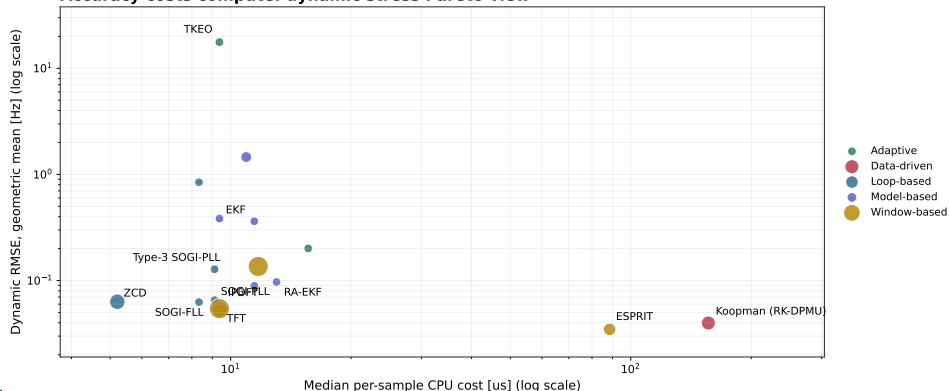
The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

Accuracy costs 100–1000× more compute

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

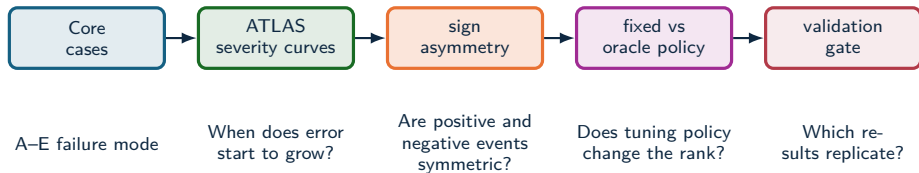
Deployable methods sit lower-left; spectral/data-driven accuracy moves right by 100–1000× CPU.

Accuracy costs compute: dynamic stress Pareto view



ATLAS severity analysis

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps — Prony stress-corner failures flagged



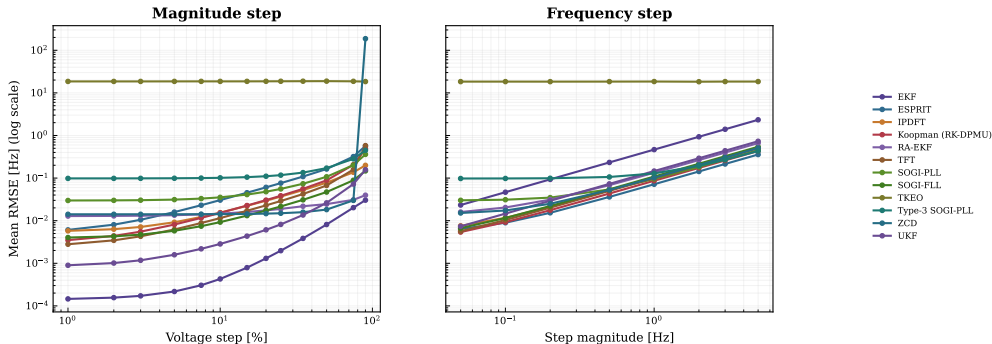
ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Severity response: amplitude and frequency steps

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps — Prony stress-corner failures flagged

Log-log RMSE vs. severity ($n = 30$). Most methods rise smoothly toward the Cramér–Rao floor; saturated curves stay high regardless of severity.

Severity response curves: RMSE grows with disturbance severity

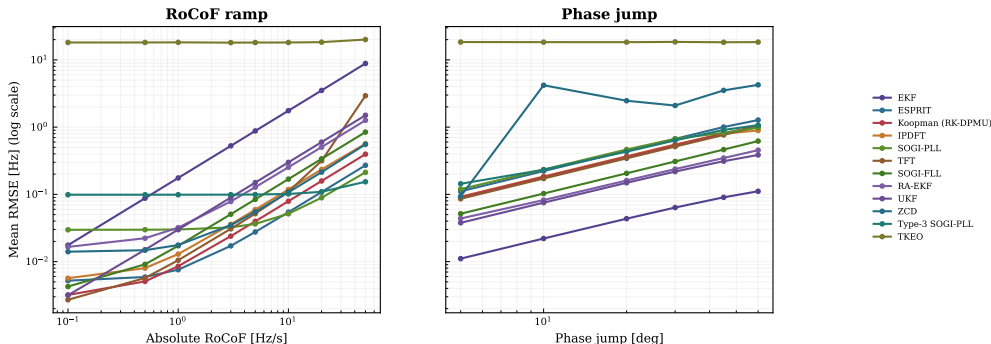


Severity response: RoCoF ramp and phase jump

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps — Prony stress-corner failures flagged

Error grows with RoCoF magnitude and phase-jump angle; the curve slope and onset are the family-specific failure signatures.

Severity response curves: RMSE grows with disturbance severity

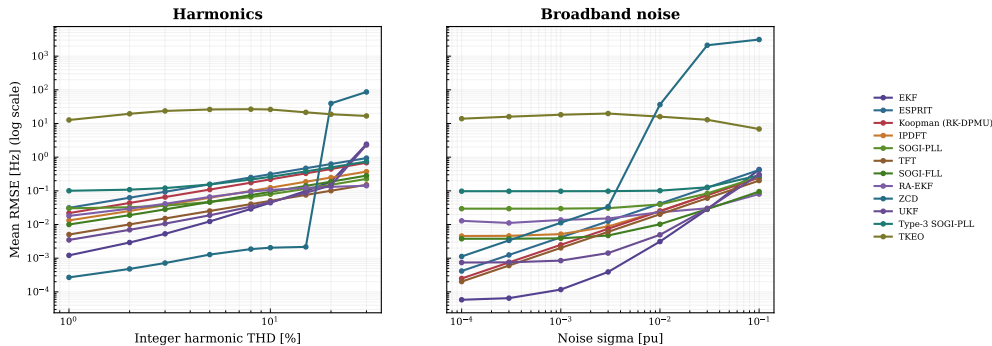


Severity response: harmonics and broadband noise

ATLAS EXTENDED SWEEP: $n = 30$, fixed policy, 9/9 sweeps — Prony stress-corner failures flagged

Spectral and noise stress: several methods hold a low-error plateau, then cross a cliff into the unusable regime as severity rises.

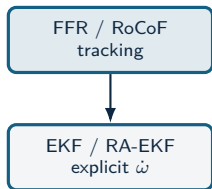
Severity response curves: RMSE grows with disturbance severity



Deployment & Conclusion

Recommendations and validation agenda

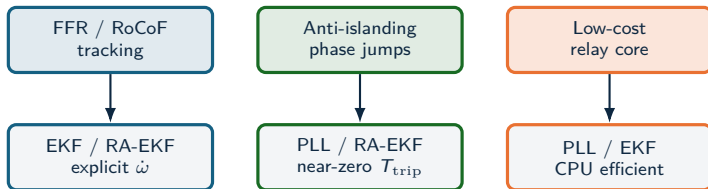
Recommendation map



Recommendation map



Recommendation map



Recommendation map



Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

Three findings

1. Metric-conditioned

The MC benchmark and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

2. Stress-bounded

AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy

Low point error does not imply robustness: RA-EKF stays within band on every ramp seed in the tested run.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Conclusion and roadmap

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Future work — roadmap

- **Phase 1:** ANDES transient simulation of IEEE IBR standard-method fault scenarios.
- **Phase 2:** Load the simulation CSVs as benchmark truth and replay every estimator.
- **Phase 3:** HIL with FPGA estimator implementations for real-time validation.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.

Selected references

Standards, measurement & methods

- IEEE/IEC 60255-118-1:2018 — synchrophasor, frequency, and RoCoF measurement and compliance requirements.
- H. Ahmed, S. Biricik, M. Benbouzid, “Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation,” IEEE, 2021 (*LKF2*).
- Open synchrophasor infrastructure: OpenPMU, openPDC, FNET/GridEye, and LBNL/PNNL open PMU event datasets.

IBR field-event reports

- NERC/WECC Blue Cut Fire (2017) and Canyon 2 Fire (2018) solar-PV interruption reports.
- NERC Odessa I/II disturbance reports (2021, 2022).
- NERC/WECC 2022 California BESS disturbance report (2023).
- Dong et al., Kaua’i 18–20 Hz oscillation analysis (arXiv:2301.05781).

Full reference list, methodology, and artifacts in the submitted paper.

Thank you

Questions & discussion welcome

Jorge L. Mayorga Taborda | Universidad de los Andes | SGSMA 2026